Declaration

```
T* fetch_add(
    ptrdiff_t i,memory_order order = memory_order_seq_cst)
    volatile noexcept;
T* fetch_add(
    ptrdiff_t i,memory_order order = memory_order_seq_cst)
noexcept;
```

Effects

Atomically retrieves the existing value of `*this` and stores *old-value* `+ i` in `*this`.

Returns

The value of `*this` immediately prior to the store.

Throws

Nothing.

Note

This is an atomic read-modify-write operation for the memory location comprising `*this`.

Std::Atomic_Fetch_Add Nonmember Function

Atomically reads the value from an `atomic<T*>` instance and replaces it with that value plus the supplied value `i` using standard pointer arithmetic rules.

Declaration

```
T* atomic_fetch_add(volatile atomic<T*>* p, ptrdiff_t i) noexcept;
T* atomic_fetch_add(atomic<T*>* p, ptrdiff_t i) noexcept;
```